Resources, Environment and Sustainability 7 (2022) 100051

Contents lists available at ScienceDirect

Resources, Environment and Sustainability

journal homepage: www.elsevier.com/locate/resenv

Benefits of biochar, poultry manure and biochar–poultry manure for improvement of soil properties and sweet potato productivity in degraded tropical agricultural soils

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ARTICLE INFO

Keywords:
Biochar
Poultry manure
Soil properties
Sweet potato
Alfisols

ABSTRACT

Studies on the effects of biochar and poultry manure on soil properties and sweet potato productivity are scarce. Hence, field experiments were conducted during 2019 and 2020 cropping seasons at two sites to evaluate the effects of biochar (B), poultry manure (PM) and their combined application on soil physical and chemical properties, growth and tuber yield of sweet potato (*Ipomoea batatas* L.). Each year, the experiment consisted of 4 × 3 factorial combinations of biochar (0, 10.0, 20.0 and 30.0 t ha^{-1}) and poultry manure (0, 5.0 and 10.0 t ha^{-1}). The experiment was laid out in a randomized complete block design with three replications. The results showed that combined application of B and PM improved soil physical and chemical properties, growth and tuber yield of sweet potato better than sole application of B and PM. In both years, there were significant interactions of B and PM (B × PM) on soil properties and sweet potato variables that were measured, indicating the potential of B in improving nutrient use efficiency of PM amendment. It was found that the highest application rate of 30.0 t ha^{-1} B and 10.0 t ha^{-1} PM (B_{30}+PM_{10}) gave the highest tuber yield of sweet potato at both sites. Averaged over two years, the highest application rate of B at 30.0 t ha^{-1} and PM at 10.0 t ha^{-1} (B_{30}+PM_{10}) significantly increased tuber yield of sweet potato by 108, 147, 196, 89, 124 and 220%, compared with sole B at 30.0 t ha^{-1} (B_{30}+PM_0), sole B at 20.0 t ha^{-1}(B_{20}+PM_0), sole B at 10.0 t ha^{-1} (B_{10}+PM_0), sole PM at 10.0 t ha^{-1}(B_0+PM_{10}), sole PM at 5.0 t ha^{-1}(B_0+PM_5) and no application of B or PM (control) (B_0+PM_0), respectively. Lowest tuber yield was obtained for the control. The multiple regressions revealed that bulk density, porosity and moisture content and soil chemical properties (pH, OC, N, P, K, Ca and Mg) significantly influenced the tuber yield of sweet potato and therefore dictated the performance of sweet potato in Alfisols of southwest Nigeria. The highest application rate of 30.0 t ha^{-1} B and 10.0 t ha^{-1} PM were found best on most of the measured soil physical and chemical properties, growth and tuber yield of sweet potato. The results indicated that B in combination with PM have the potential to improve soil properties and sweet potato yield of severely degraded soils.

1. Introduction

Sweet potato (*Ipomoea batatas* L.) is produced abundantly in the tropics (Agbede, 2010). The tuber is eaten boiled, fried and baked or made into syrup and also serves as raw materials for the manufacture of starch, glucose and alcohol (Yasmin et al., 2007). The leaves are also used as vegetables in some communities in Nigeria and widely used as fodder for livestock. The tuber is an excellent source of anti-oxidants and carotenes chiefly in the orange flesh colour varieties thus providing cheap and rich source of vitamin A for poor people (Woolfe, 1992). However, the average sweet potato productivity per hectare on farmlands in the tropics is very low, and has been attributed mostly to poor soil conditions. According to Zingore et al. (2003) and Agbede (2010), tropical soils are low in organic matter and available nutrients and for this reason productivity and sustainability decline over time. This is due to continuous cultivation of the same land, which leads to soil nutrient depletion, physical degradation and reduced yield which in turn calls for enhanced crop productivity through application of soil amendments to correct the anomalies or reverse the negative nutrients and maintain productivity. Therefore the replenishment of nutrients and enhanced quality of tropical soils could be achieved through the addition of organic matters such as, manures, mulches and composts. According to Osundare (2004), sweet potato like any other tuber crop is a heavy feeder exploiting greater volume of soil for

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https://doi.org/10.1016/j.resenv.2022.100051
Received 14 April 2021; Received in revised form 12 February 2022; Accepted 12 February 2022
Available online 22 February 2022
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nutrients and water. Study into sustainable soil management package for enhanced productivity of sweet potato is rare. Therefore, the fragile nature of tropical soils required the appropriate agronomic practices and integrated soil management for sustainable use of soils and high crop productivity.

Among agricultural soil amendments that can enhance agricultural productivity and improve soil quality and sustainability is biochar (Lehmann and Joseph, 2015; Guo et al., 2016; Igalavithana et al., 2016). Biochar is the product of pyrolysis of organic materials in the absence of oxygen and at high temperature. The technique of using charcoal (biochar) to improve the fertility of soils originated in the Amazon Basin of South America for more than 2500 years (Lehmann et al., 2003; Semida et al., 2019); United States Biochar Initiative-http://biochar-us.org/biochar-then-now. The native Indians of the region would create charcoal (biochar) and incorporate it in small plots of land from 1–80 hectares in size (http://www.isleofwightwoodlandproducts.co.uk/biochar). Dark, charcoal-rich soil (known as Terra Preta or black earth) as it is known in this area of Brazil, remains highly fertile until today, even with little or no application of fertilizers. And this is in a region of the world known for its highly infertile soils. Terra Preta supported productive farms in areas that previously had poor, and in some places, toxic soils (United States Biochar Initiative-). Before now organic matters, such as manures, mulches and composts were being used to improve the fertility or productivity of the soil. However, the benefits of organic matters are generally short-lived because they decomposed rapidly, especially under tropical conditions with high temperature and aeration (Mekuria and Noble, 2013). Therefore, the use of more stable compounds, such as biochar may be one of the soil management options. There are two aspects which make biochar amendment superior to other organic materials: the first is the high stability against decomposition, in order that it can stay in soil for longer times providing long-term benefits to soil and the second is having extra capability to retain the nutrients. Biochar amendment improves soil quality by increasing soil pH, moisture-holding capacity, cation-exchange capacity, and microbial flora (Mensah and Frimpong, 2018). Studies have shown that biochar application improved the physical, chemical and biological properties and therefore crop yield. Although biochar recently has attracted considerable interest as a sustainable technology to improve soil fertility in the tropics, information on their potential to amend degraded sandy soil and sandy loam soil under sweet potato cropping system in the field conditions in the humid tropics of southwestern Nigeria does not exist. Where such studies were investigated, they were pot experiments in a greenhouse (Fagbenro et al., 2013; Yusif et al., 2018; Olakayode et al., 2019). Few field studies have documented the influence of biochar and organic manure on soil fertility and other soil properties in Nigeria (Are et al., 2017; Adekiya et al., 2019) and other tropical countries (Arif et al., 2012; Agegnehu et al., 2016; Mensah and Frimpong, 2018). Furthermore, most of the previous studies on positive impacts of biochar on soil health, carbon sequestration, soil fertility improvement, hydrological properties, soil physical quality, reduction of greenhouse gas emission and global warming, reduction of soil nutrient leaching losses, and crop productivity on degraded soils, were focused on temperate soils (Schulz et al., 2013; Basso et al., 2013; Głąb et al., 2016; Oni et al., 2019). Whereas the mechanism explaining its impact on severely degraded soil in sub-Saharan Africa is still not adequately investigated. While biochar represents a soil conditioner and valuable soil ameliorant that can change the physical and chemical properties of the soil, it might be limited as a nutrient supplier alone, because of its very low nutrient concentrations and high recalcitrant to biodegradation (Partey et al., 2014). Therefore, for improved soil sustainability and sweet potato productivity, application of poultry manure in combination with biochar may be one of the best options. Poultry manure, which originates from the decomposition of chicken droppings, is considered as a valuable organic manure (Agbede et al., 2017; Adekiya et al., 2019). Poultry manure is high in organic materials and contains nutrients essential for crop production. The application of poultry manure as organic fertilizers remains famous popular globally and agricultural use of poultry manure is found to have a definite advantage to the soil quality improvement. The application of poultry manure to the soil increased soil organic matter and other plant nutrients, thereby positively improved soil physical and chemical properties, and crop yields (Agbede and Ojeniyi, 2009; Abd El-Kader et al., 2010; Agbede et al., 2017; Adekiya et al., 2019).

In Nigeria, few field studies had been investigated to evaluate the effects of biochar application to soil in combination with organic fertilizer or inorganic fertilizer on crop yield. Dou et al. (2012) and Chan et al. (2007) reported that biochar application to soils in combination with either organic or inorganic fertilizer have a remarkable impact on plant growth and yield. For higher crop yields to be obtained from physically degraded soils, Revell et al. (2012) and Naeem et al. (2017) advocated the application of biochar in combination with other organic residues such as compost or manure. Use of such soil amendments for improving soil quality as well as crop productivity in the forest-savanna transition zone of southwest Nigeria is rare. The working hypothesis in this study was that the combined application of biochar and poultry manure would significantly improve soil physical and chemical properties, growth and sweet potato yield compared with the sole application of the biochar and poultry manure. Therefore, the objective of this study was to determine the effects of sole biochar and poultry manure and their combined application at different rates on soil physical and chemical properties, growth and yield of sweet potato in degraded sandy soil and sandy loam soil under the forest-savanna transition zone of southwest Nigeria.

2. Materials and methods

2.1. Site description and treatments

Field experiments were carried out at the Teaching and Research Farm of Rufus Giwa Polytechnic (Site A - 7°13′29.0″N–7°13′30.9″N and longitude 5°32′52.3″E–5°32′54.2″E, with elevation varying from 314 to 320 m above sea level) and Obasooto village (Site B - 7°12′30.5″N–7°12′31.8″N and longitude 5°32′53.3″E–5°32′54.1″E, with elevation varying from 272 to 280 m above sea level) in Owo, Ondo State, Nigeria during the 2019 and 2020 cropping seasons. Obasooto is located in the west of the Owo region and is about 10 km from Owo. Both sites are located within the forest-savanna transition zone of southwest Nigeria. The soils at Owo and Obasooto have a basement complex texture belonging to the Alfisols, classified as Oxic Tropuldalf (Soil Survey Staff, 2014) or Luvisol (IUSS Working Group WRB, 2015) and locally classified as Okemesi Series (Smyth and Montgomery, 1962), derived from quartzite, gneiss and schist. The physical and chemical properties of the soils at site A and site B before the commencement of the experiment are presented in Table 1. The soil at site A was purely sand in texture, while soil at site B was sandy loam in texture (Table 1). The soils at both sites had high bulk density and moderately acidic, and were generally low in organic carbon (OC), total N, available P, exchangeable K, Ca and Mg. This could be attributed to soil degradation as result of continuous arable cropping of the soils of sites of experimentation. Average annual precipitation is about 1400 mm, and mean annual temperature is about 32 °C. The sites were under a year fallowed after rotational arable cropping, and none of the sites had received fertilizer application during the previous 6 years. Previous soil management included tillage practice such as conventional tillage system which involves ploughing, harrowing, and ridging.

The experiment consisted of 4 × 3 factorial combinations of biochar (B) (0, 10.0, 20.0 and 30.0 t ha^{-1}) and poultry manure (PM) (0, 5.0 and 10.0 t ha^{-1}). The twelve treatments were factorially arranged in a randomized complete block design with three replications. Each block comprised of 12 plots and each plot was 5 × 4 m. Blocks were 1 m

Table 1
Soil physical and chemical properties (0–15 cm depth) of the site A and site B prior to experimentation in 2019.

Property	Site A	Site B
Sand (%)	92	76
Silt (%)	3	13
Clay (%)	5	11
Textural class	Sand	Sandy loam
Bulk density (Mg m^{-3})	1.61	1.58
pH (water)	5.51	5.52
Organic carbon (%)	1.23	1.34
Total N (%)	0.12	0.14
Available P (mg kg^{-1})	6.75	8.12
Exchangeable K (cmol kg^{-1})	0.11	0.12
Exchangeable Ca (cmol kg^{-1})	1.35	1.51
Exchangeable Mg (cmol kg^{-1})	0.37	0.39

Table 2
Chemical composition of biochar and poultry manure used in the experiment.

Property	Biochar	Poultry manure
pH (water)	7.86	6.25
Ash (%)	8.32	12.1
Organic C (%)	55.7	22.3
Nitrogen (%)	0.85	2.89
C/N	65.5	7.72
Phosphorous (%)	0.38	1.34
Potassium (%)	1.92	1.57
Calcium (%)	4.63	0.92
Magnesium (%)	3.78	0.46
Copper (%)	0.013	0.37
Manganese (%)	0.068	0.21
Sulphur (%)	0.10	0.33
Zinc (%)	0.008	0.24
Sodium (%)	0.21	0.27

apart and plots were 0.5 m apart. Crop establishments were carried out in April each year. The same location was used at each site for the two years of the experiment. The amendments were applied twice (one application each year).

2.2. Preparation and analysis of biochar and poultry manure

Biochar used in the experiment was obtained from a nearby industrial charcoal producer at Owo, Ondo State, Nigeria who makes use of hardwood such as *Parkis biglosa, Khaya senegalensis, Prosopis africana* and *Terminalia glaucescens* in traditional kilns to produce charcoal for domestic use. The temperature inside the kiln was monitored with a thermocouple thermometer and had an average temperature of 580 °C for 24 h of carbonizing. The biochar was ground and sieved through a 2 mm sieve before application. The poultry manure (PM) was obtained from the poultry unit of the Teaching and Research Farm of Rufus Giwa Polytechnic, Owo, Ondo State. The poultry manure droppings were collected from layer chickens, egg laying breed (Leghorn) which were reared/kept in battery cages. The egg laying chickens were fed with Top Feeds Layer Mash and kept for 12 months from their first laying period. The poultry manure was composted for 3 weeks to allow for mineralization (Adekiya et al., 2019).

The chemical analysis of the biochar and poultry manure used for the experiments were determined following standard procedures (Tel and Hagarty, 1984). The biochar was alkaline, while poultry manure was slightly acidic in nature. Biochar had higher concentrations of OC, K, Ca, Mg and a high C:N ratio compared with poultry manure, but poultry manure had higher concentrations of *N* and P and micronutrients compared with biochar (Table 2).

2.3. Land preparation, incorporation of biochar and poultry manure and planting of sweet potato vines

The sites were manually cleared and weeds removed. Thereafter, the experimental sites were laid out to the required plot size of 5 × 4 m. The soils were then disked to a 20 cm depth using a hand held hoe. The biochar (B) and poultry manure (PM) were weighed and spread evenly on the plots according to the required rates (B: 0, 10.0, 20.0 and 30.0 t ha^{-1}; PM; 0, 5.0 and 10.0 t ha^{-1}) over the soil. This study used biochar application rate within the recommended levels (5–50 t ha^{-1}) by the International Biochar Initiative (Jirka and Tomlinson, 2015). The poultry manure rate (5–10 t ha^{-1}) used for this study was adopted from Agbede (2010), while the treatment combination used the integrated plant nutrition system strategy (IPNS), i.e. combining full of the sole application of the biochar at different levels with half and full of the sole application of the poultry manure. A hand held hoe was used to incorporate the amendments into the soil to the depth of approximately 10 cm. The biochar and poultry manure were incorporated into the soil on 1st April each year and planting of sweet potato vines was done 2 weeks after incorporation of the amendments into the soil in order to allow equilibration of the amendments in the soil and for mineralization to start. The hardwood feedstock and poultry manure were selected based on their accessibility, availability and sustainability in the region.

Sweet potato (*Ipoemea batatas* L. local variety) vines about 40 cm long were planted in April each year of the experiment after soil disking. One sweet potato vine was planted per hole at a spacing of 1 m × 1 m, giving sweet potato population of 10,000 plants ha^{-1}. Weeding was done manually twice at 3 and 8 weeks after planting (DAP). No irrigation water was performed during the course of the experiment.

2.4. Determination of soil physical and chemical properties

Some soil physical properties were determined in all plots at 2, 3, 4 and 5 months after planting sweet potato. Five samples were collected using a cylindrical steel core (4 cm diameter, 15 cm high) at 0–15 cm depth from the centre of each plot at random and about 15 cm away from each sweet potato stand. The samples were used to evaluate bulk density, total porosity and gravimetric moisture content after oven-drying at 100 °C for 24 h. Total porosity was calculated from the values of bulk density and particle density of 2.65 Mg m^{-3}.

Before the commencement of the experiment in 2019, soil samples were taken from 0–15 cm depth at 10 different points selected randomly from the experimental site. Disturbed soil samples were also collected randomly at 0–15 cm depth from the centre of each plot at five sites per plot at harvest in 2019 and 2020. The soil samples collected were bulked, air-dried and sieved using a 2-mm sieve for routine chemical analysis, as described by Carter and Gregorich (2007). Particle-size analysis was done using the hydrometer method. Textural class was determined using a textural triangle. Soil pH was determined in a soil/water (1:2) suspension using a digital electronic pH metre. Soil organic carbon content was determined by the Walkley and Black procedure using dichromate wet oxidation. Total *N* content was determined using micro-Kjeldahl digestion and distillation techniques. The content of available P was determined by Bray-1 extraction followed by molybdenum blue colorimetry. Exchangeable K, Ca and Mg were extracted with a 1 M ammonium acetate (NH_4OA_C), pH 7 solution. Thereafter, exchangeable K was analysed with a flame photometer, and exchangeable Ca and Mg were determined with an atomic absorption spectrophotometer.

2.5. Determination of crop growth and yield parameters

Ten plants were selected randomly per plot for determination of vine length and leaf area at 90 days after planting (DAP) when the sweet potato plant reached its peak growth. Vine length was measured by metre rule. Leaf area was measured by graphical method (i.e. by placing the leaf on graph sheet for area determination). Yield attributes measured included number of tubers, tuber weight (kg $plant^{-1}$) and tuber yield (t ha^{-1}). These were measured at 5 months after planting (MAP) by harvesting ten sweet potato plants per plot. The total number

Table 3
Effect of sites, biochar and poultry manure on soil physical properties (0–15 cm depth) when averaged across four sampling periods (2, 3, 4 and 5 months after planting).

Site	Biochar (t ha^{-1})	Poultry manure (t ha^{-1})	Bulk density (Mg m^{-3})		Porosity (%)		Moisture content (%)	
			2019	2020	2019	2020	2019	2020
Site A	0.0	0.0	1.68	1.69	36.6	36.2	10.6	11.5
	0.0	5.0	1.55	1.53	41.5	42.3	11.9	12.8
	0.0	10.0	1.44	1.42	45.7	46.4	13.0	13.9
	10.0	0.0	1.45	1.43	45.3	46.0	11.6	12.5
	10.0	5.0	1.36	1.34	48.7	49.4	15.3	16.2
	10.0	10.0	1.23	1.21	53.6	54.3	18.0	18.9
	20.0	0.0	1.24	1.22	53.2	54.0	16.6	17.5
	20.0	5.0	1.13	1.11	57.4	58.1	18.4	19.3
	20.0	10.0	1.02	1.01	61.5	61.9	20.9	21.8
	30.0	0.0	1.04	1.13	60.8	57.4	19.5	20.4
	30.0	5.0	0.96	0.94	63.8	64.5	21.7	22.6
	30.0	10.0	0.87	0.85	67.2	67.9	24.0	24.9
Site B	0.0	0.0	1.57	1.55	40.8	41.5	11.8	13.6
	0.0	5.0	1.45	1.43	45.3	46.0	13.1	14.4
	0.0	10.0	1.35	1.33	49.1	49.8	14.2	15.5
	10.0	0.0	1.36	1.34	48.7	49.4	12.8	14.1
	10.0	5.0	1.27	1.25	52.1	52.8	16.5	17.8
	10.0	10.0	1.16	1.14	56.2	57.0	19.2	20.5
	20.0	0.0	1.17	1.15	55.8	56.6	17.8	19.1
	20.0	5.0	1.06	1.04	60.0	60.8	19.6	20.9
	20.0	10.0	0.99	0.97	62.6	63.4	22.1	23.3
	30.0	0.0	1.00	0.99	62.3	62.6	20.7	22.0
	30.0	5.0	0.92	0.90	65.3	66.0	22.9	24.2
	30.0	10.0	0.81	0.79	69.4	70.2	25.2	26.5
		LSD (0.05)	0.11	0.09	4.2	3.9	0.5	0.4
Site (S)			0.000	0.000	0.000	0.000	0.000	0.000
Biochar (B)			0.000	0.000	0.000	0.000	0.000	0.000
Poultry manure (PM)			0.000	0.000	0.000	0.000	0.000	0.000
S × B			0.001	0.000	0.006	0.007	0.023	0.034
S × PM			0.007	0.000	0.009	0.006	0.043	0.009
B × PM			0.003	0.000	0.004	0.007	0.000	0.000
S × B × PM			0.764	0.081	1.000	0.994	1.000	1.000

of tubers produced by each plant were physically counted and recorded as the number of tubers; their weights were determined and recorded as the tuber weight and thereafter converted to tuber yield in tons per hectare.

2.6. Statistical analysis

The experiments were conducted in a randomized complete block design, with factorial arrangements so that the main effects of site (S), biochar (B) and poultry manure (PM) and the interactions of S × B, S × PM, B × PM and S × B × PM on soil physical and chemical properties, growth and yield of sweet potato could be tested. The data collected were analysed using two-way analysis of variance (ANOVA) with the Genstat for Windows 21st Edition software package (Genstat, 2021). Means were separated using Fisher's least significant difference (LSD) test at the 0.05 probability level. Multiple regressions were calculated to determine how the soil properties are related to the leaf area and yield.

3. Results

3.1. Effect of site, biochar and poultry manure on soil physical properties

The effect of site and the applications of biochar and poultry manure on soil physical properties 2019 and 2020 are presented in Table 3. In both years (2019 and 2020), site, biochar and poultry manure played a significant role in influencing the physical properties of the soil when studied as individual factors. In both years (2019 and 2020), biochar and poultry manure treatments significantly reduced bulk density, and increased porosity and moisture content compared with the control. Moreover, the bulk density decreased, while the porosity and moisture content increased with the rate of biochar and poultry manure application. In both years and when studied as individual factor, application of biochar significantly ($p = 0.05$) influenced soil physical properties — reduced bulk density, and increased porosity and moisture content compared with the control. Similarly, poultry manure as individual factor also significantly improved soil physical properties compared with the control. In both years, the interactive effect of site (S) and biochar (B), site (S) and poultry manure (PM) and B × PM for soil physical properties were significant. In both years, when all three factors (S × B × PM) were considered together, interactions were not significant.

3.2. Effect of site, biochar and poultry manure on soil chemical properties

The effect of site, biochar and poultry manure on soil chemical properties in 2019 and 2020 are shown in Table 4. In both years (2019 and 2020), site, biochar and poultry manure played a significant role influencing the soil chemical properties when studied as individual factors. In both years (2019 and 2020), biochar or poultry manure applied alone significantly increased soil pH, organic carbon (OC), N, P, K, Ca and Mg, and concentration increased with increasing biochar and poultry manure application rates. The biochar increased soil pH, OC, N, P, K, Ca and Mg with rate of biochar from 0 to 30.0 t ha^{-1}. Similarly, poultry manure increased soil pH, OC, N, P, K, Ca and Mg with rate of poultry manure from 0 to 10.0 t ha^{-1}. The application rate of 30.0 t ha^{-1} biochar + 10.0 t ha^{-1} poultry manure (B_{30} + PM_{10}) gave the highest soil chemical properties among all the treatments. The control (no application of biochar or poultry manure) had the least values of soil chemical properties. In both years and when studied as individual factor, site significantly ($p = 0.05$) influenced concentrations of soil pH, OC, N, P, K, Ca and Mg. When studied as individual factor, biochar significantly ($p = 0.05$) increased concentrations of soil pH, OC, N, P, K, Ca and Mg. Likewise, the application of poultry manure as individual factor, significantly ($p = 0.05$) increased soil chemical properties in

Table 4
Effect of sites, biochar and poultry manure on soil chemical properties (0–15 cm depth).

Site	Biochar (t ha^{-1})	Poultry manure (t ha^{-1})	pH (water)		OC (%)		Total N (%)		Available P (mg kg^{-1})		Exchangeable K (cmol kg^{-1})		Exchangeable Ca (cmol kg^{-1})		Exchangeable Mg (cmol kg^{-1})	
			2019	2020	2019	2020	2019	2020	2019	2020	2019	2020	2019	2020	2019	2020
Site A	0.0	0.0	5.51	5.49	1.14	1.02	0.11	0.10	6.5	5.3	0.09	0.08	1.21	1.16	0.34	0.32
	0.0	5.0	5.96	6.18	1.32	1.43	0.13	0.15	8.6	10.9	0.11	0.12	1.36	1.41	0.54	0.59
	0.0	10.0	6.18	6.41	1.54	1.66	0.15	0.17	10.8	13.5	0.13	0.14	1.51	1.56	0.62	0.67
	10.0	0.0	6.24	6.47	1.75	1.86	0.13	0.14	7.2	8.4	0.14	0.16	1.66	1.71	0.71	0.76
	10.0	5.0	6.40	6.63	1.81	1.93	0.17	0.19	12.8	15.5	0.16	0.18	1.82	1.88	0.65	0.71
	10.0	10.0	6.51	6.74	1.93	2.11	0.19	0.21	14.9	17.9	0.20	0.22	1.98	2.05	0.81	0.88
	20.0	0.0	6.54	6.77	1.87	1.98	0.17	0.18	8.4	9.6	0.18	0.20	2.11	2.16	0.79	0.84
	20.0	5.0	6.65	6.86	2.11	2.25	0.21	0.23	17.5	20.5	0.23	0.24	2.27	2.34	0.89	0.93
	20.0	10.0	6.70	6.93	2.29	2.44	0.23	0.25	19.8	22.8	0.27	0.28	2.43	2.50	0.97	1.01
	30.0	0.0	6.78	7.01	2.24	2.35	0.19	0.20	9.6	10.3	0.30	0.32	2.56	2.61	1.01	1.07
	30.0	5.0	6.89	7.12	2.57	2.83	0.25	0.27	21.4	25.4	0.34	0.36	2.72	2.78	1.12	1.19
	30.0	10.0	6.97	7.20	2.85	3.12	0.27	0.29	24.1	27.7	0.38	0.41	2.88	2.95	1.21	1.13
Site B	0.0	0.0	5.52	5.51	1.25	1.14	0.13	0.12	7.2	7.2	0.10	0.09	2.21	2.08	0.36	0.37
	0.0	5.0	6.01	6.23	1.43	1.62	0.15	0.17	9.7	9.7	0.12	0.13	2.49	2.54	0.66	0.69
	0.0	10.0	6.23	6.46	1.61	1.81	0.17	0.19	12.3	12.3	0.14	0.16	2.78	2.83	0.74	0.77
	10.0	0.0	6.29	6.52	1.72	1.99	0.15	0.16	7.9	6.9	0.15	0.17	3.06	3.10	0.83	0.86
	10.0	5.0	6.45	6.68	1.90	2.18	0.20	0.22	14.8	16.1	0.17	0.19	3.35	3.40	0.72	0.75
	10.0	10.0	6.56	6.79	2.08	2.37	0.22	0.24	17.6	18.8	0.21	0.23	3.64	3.69	0.93	0.96
	20.0	0.0	6.59	6.82	2.21	2.53	0.19	0.20	8.6	8.9	0.19	0.21	3.92	3.96	0.91	0.93
	20.0	5.0	6.68	6.91	2.39	2.74	0.25	0.27	20.3	21.1	0.24	0.26	4.21	4.26	1.01	1.04
	20.0	10.0	6.75	6.98	2.57	2.95	0.27	0.29	22.9	24.0	0.28	0.30	4.50	4.55	1.09	1.11
	30.0	0.0	6.83	7.06	2.72	3.11	0.20	0.21	9.3	9.7	0.31	0.33	4.79	4.83	1.13	1.16
	30.0	5.0	6.94	7.17	2.91	3.34	0.29	0.31	25.5	26.4	0.35	0.37	5.08	5.14	1.25	1.28
	30.0	10.0	7.02	7.25	3.08	3.59	0.31	0.33	28.2	29.6	0.39	0.42	5.39	5.45	1.37	1.40
		LSD (0.05)	0.28	0.25	0.06	0.05	0.005	0.004	0.31	0.29	0.004	0.003	0.06	0.04	0.01	0.009
Site (S)			0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.005	0.001	0.000	0.000	0.000	0.000	0.000
Biochar (B)			0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Poultry manure (PM)			0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
S × B			0.000	0.000	0.000	0.000	0.042	0.041	0.034	0.009	0.008	0.034	0.000	0.000	0.005	0.000
S × PM			0.000	0.000	0.000	0.003	0.002	0.024	0.000	0.005	0.006	0.009	0.000	0.000	0.021	0.000
B × PM			0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.017	0.000	0.000	0.000
S × B × PM			0.000	0.000	0.000	0.000	0.000	0.002	0.551	0.120	0.000	0.000	0.000	0.000	0.003	0.000

both years. In both years, the interactive effect of S × B, S × PM and B × PM were significant for soil pH, OC, N, P, K, Ca and Mg. When all three factors (S × B × PM) were considered together, interactions were significant for soil pH, OC, N, K, Ca and Mg, but interactions was not significant for soil P in both years.

3.3. Effect of site, biochar and poultry manure on growth and tuber yield of sweet potato

The effect of site, biochar and poultry manure on the vine length, leaf area and tuber yield of sweet potato in 2019 and 2020 are presented in Table 5. In both years (2019 and 2020), site, biochar and poultry manure significantly influenced growth and tuber yield of sweet potato. In the first and second year, biochar significantly ($p = 0.05$) increased vine length and leaf area (Fig. 1, Table 5) and tuber yield (Fig. 2, Table 5) of sweet potato with the rate of application from 0 to 30.0 t ha^{-1}. Similarly, poultry manure significantly ($p = 0.05$) increased vine length and leaf area (Fig. 1, Table 5) and tuber yield (Fig. 2, Table 5) of sweet potato with the rate of application from 0 to 10.0 t ha^{-1}. In both years, the highest application rate of 30.0 t ha^{-1} biochar + 10.0 t ha^{-1} poultry manure gave the highest vine length, leaf area and tuber yield of sweet potato compared with all other treatments. The control gave the least values of vine length, leaf area and tuber yield of sweet potato in both years. When studied as individual factor, site influenced vine length, leaf area and tuber yield of sweet potato significantly in both years (Table 5). Biochar and poultry manure as individual factors also affected vine length, leaf area and tuber yield of sweet potato in both years. When biochar and poultry manure (B × PM) are considered, the interaction was significant for vine length, leaf area and tuber yield of sweet potato in both years. However, the interactive effect of S × B and S × PM for vine length, leaf area and tuber yield of sweet potato were not significant, nor was the S × B × PM interaction in both years.

Therefore, at site A and using the mean of the first and second years (2019 and 2020), $B_0 + PM_0$, $B_0 + PM_5$, $B_0 + PM_{10}$, $B_{10} + PM_0$, $B_{10} + PM_5$, $B_{10} + PM_{10}$, $B_{20} + PM_0$, $B_{20} + PM_5$, $B_{20} + PM_{10}$, $B_{30} + PM_0$, $B_{30} + PM_5$ and $B_{30} + PM_{10}$ treatments, produced 10.1, 15.0, 17.7, 10.8, 20.4, 23.2, 13.1, 25.9, 28.7, 15.7, 31.3 and 34.0 t ha^{-1}, respectively, for sweet potato tuber yield. The corresponding sweet potato tuber yield at site B were 12.3, 17.1, 20.2, 13.4, 21.8, 25.1, 15.9, 26.5, 30.0, 18.6, 34.4 and 37.6 t ha^{-1}, respectively. Pooled across years, site, biochar and poultry manure treatments; $B_0 + PM_0$, $B_0 + PM_5$, $B_0 + PM_{10}$, $B_{10} + PM_0$, $B_{10} + PM_5$, $B_{10} + PM_{10}$, $B_{20} + PM_0$, $B_{20} + PM_5$, $B_{20} + PM_{10}$, $B_{30} + PM_0$, $B_{30} + PM_5$ and $B_{30} + PM_{10}$, produced 11.2, 16.0, 18.9, 12.1, 21.1, 24.1, 14.5, 26.2, 29.3, 17.2, 32.8 and 35.8 t ha^{-1}, respectively, for sweet potato tuber yield. Therefore, using the mean of the two year cropping seasons, application of biochar at 30.0 t ha^{-1} and poultry manure at 10.0 t ha-^{1}($B_{30} + PM_{10}$) significantly increased tuber yield of sweet potato by 108, 147, 196, 89, 124 and 220%, compared with sole biochar at 30.0 t ha^{-1} ($B_{30} + PM_0$), sole biochar at 20.0 t ha^{-1}($B_{20} + PM_0$), sole biochar at 10.0 t ha^{-1} ($B_{10} + PM_0$), sole poultry manure at 10.0 t ha^{-1}($B_0 + PM_{10}$), sole poultry manure at 5.0 t ha^{-1}($B_0 + PM_5$) and no application of biochar or poultry manure (control) ($B_0 + PM_0$), respectively.

When soil physical properties (bulk density, porosity and moisture content) were regressed as independent variables with leaf area and tuber yield as dependent variables (Table 6), the coefficient of determination (R^2) for leaf area and tuber yield were 0.869 and 0.775, respectively. The multiple regressions revealed that bulk density, porosity and moisture content significantly influenced the leaf area and tuber yield of sweet potato. When soil chemical properties (pH, OC, N, P, K, Ca and Mg) were regressed as independent variable (Table 7), the R^2 for leaf area and tuber yield were 0.984 and 0.987, respectively. The multiple regressions showed that pH, OC, N, P, K, Ca and Mg significantly influenced the leaf area and tuber yield of sweet potato.

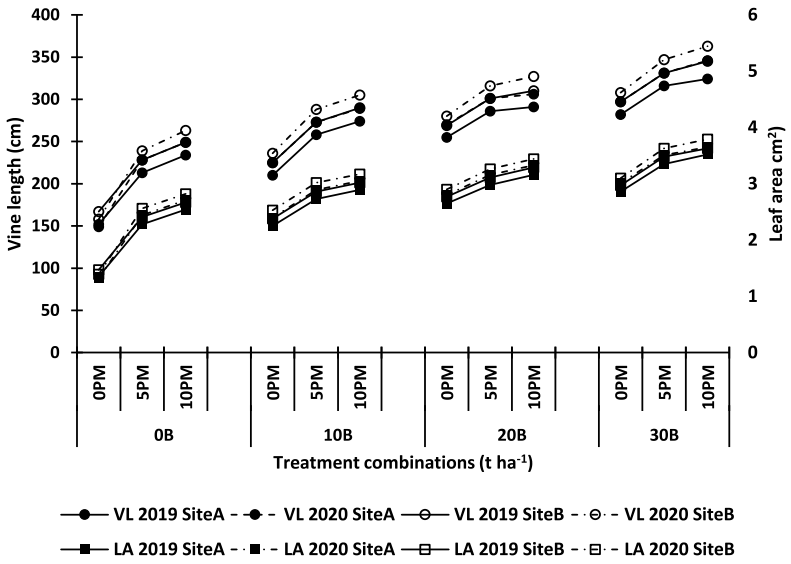


Fig. 1. Effect of site, biochar and poultry manure on vine length (LA) and leaf area (LA) of sweet potato.

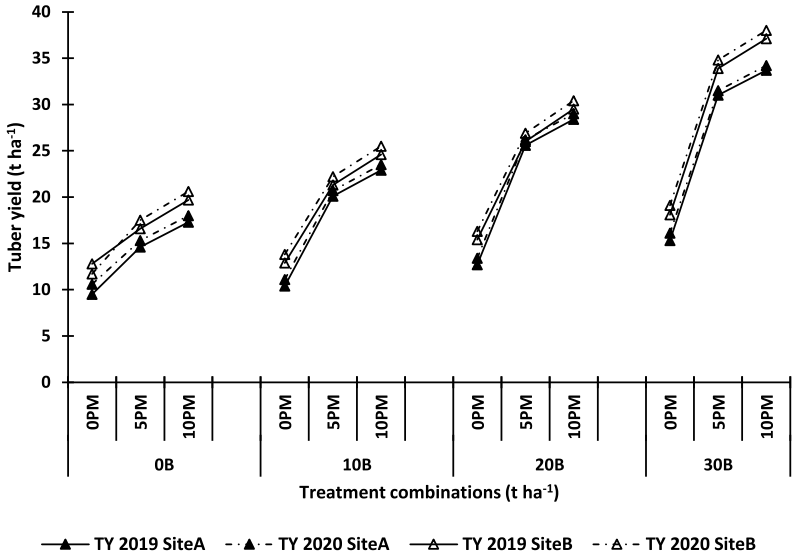


Fig. 2. Effect of site, biochar and poultry manure on tuber yield (TY) of sweet potato.

4. Discussion

Results showed that both experimental sites (A and B) had low concentrations of OC, N, P, K, Ca and Mg, and acidic with high bulk density. These conditions are the characteristics of Alfisols of southwest Nigeria (Lal, 1986; deRidder and van Keulen, 1990). The high soil bulk densities before the commencement of the experiment was attributed to the low organic matter of the sites and sandy nature of the soils. The higher bulk density values obtained in this study may be due to compaction by animal traction as common in southwestern Nigeria or continuous tractorization. The soils at both sites on which the experiments were carried out were degraded soils, on a relatively flat terrain that has been under continuous cultivation for more than 6 years without fertilization.

Biochar and poultry manure application alone, or in combination at different levels, led to reduced soil bulk density and increased moisture content and porosity, consistent with the results of previous laboratory incubation experiments based on organic amendments (Herath et al., 2013; Githinji, 2014; Gamage et al., 2016). Decreases in bulk density and increases in porosity and moisture content would have mediated soil biophysical environment for root and microbial respiration (Basso et al., 2013). Application of poultry manure has been reported to improve soil physical properties (Agbede et al., 2017). The results of this study showed that the application of biochar significantly reduced the bulk density of the sandy soil and sandy loam studied (Table 3), which is consistent with previous studies on both fine (Chaganti and Crohn, 2015) and coarse textured soils (Lim et al., 2016). The mechanisms responsible for the reduction in bulk density after biochar application may be attributed to the low bulk density of biochar (<0.6 Mg m^{-3}) than field soil (~1.2 Mg m^{-3}). Hence biochar application reduces the density of the bulk soil through the dilution or mixing effect (Lehmann et al., 2011; Alburquerque et al., 2014). Biochar has been reported to reduce bulk density by interacting with soil particles and improving aggregation and porosity (Blanco-Canqui, 2017). Lim et al. (2016) earlier demonstrated that biochar can alter the packing of soil particles, and create additional external soil porosity. The fact that overall, biochar and poultry manure application reduced bulk density, highlight their potential to improve soil structural development and stability.

Biochar application led to increased moisture content at sampling time, corroborates the findings of Głąb et al. (2016) and Zhang et al.

Table 5
Effect of sites, biochar and poultry manure on vine length and leaf area at 90 days after planting and tuber yield of sweet potato at 5 months after planting.

Site	Biochar (t ha^{-1})	Poultry manure (t ha^{-1})	Vine length (cm)		Leaf area per plant (cm^2)		Tuber yield (t ha^{-1})	
			2019	2020	2019	2020	2019	2020
Site A	0.0	0.0	152	149	1.35	1.32	9.5	10.6
	0.0	5.0	213	228	2.28	2.44	14.6	15.3
	0.0	10.0	234	249	2.54	2.71	17.3	18.0
	10.0	0.0	210	225	2.25	2.40	10.4	11.1
	10.0	5.0	258	273	2.73	2.89	20.1	20.7
	10.0	10.0	274	289	2.89	3.05	22.9	23.5
	20.0	0.0	255	270	2.65	2.80	12.7	13.4
	20.0	5.0	286	301	2.98	3.16	25.6	26.2
	20.0	10.0	291	306	3.16	3.33	28.4	29.0
	30.0	0.0	282	297	2.86	3.01	15.3	16.1
	30.0	5.0	316	331	3.35	3.51	31.0	31.5
	30.0	10.0	324	346	3.52	3.65	33.7	34.2
Site B	0.0	0.0	167	158	1.47	1.39	12.8	11.7
	0.0	5.0	228	239	2.41	2.56	16.6	17.5
	0.0	10.0	249	263	2.67	2.82	19.7	20.6
	10.0	0.0	225	236	2.38	2.53	12.9	13.8
	10.0	5.0	273	288	2.86	3.02	21.3	22.2
	10.0	10.0	290	305	3.02	3.17	24.6	25.5
	20.0	0.0	269	280	2.77	2.90	15.4	16.3
	20.0	5.0	301	316	3.11	3.26	26.0	26.9
	20.0	10.0	310	327	3.29	3.44	29.5	30.4
	30.0	0.0	297	308	2.97	3.10	18.1	19.1
	30.0	5.0	331	347	3.48	3.63	33.9	34.8
	30.0	10.0	345	363	3.63	3.79	37.1	38.0
		LSD (0.05)	13	11	0.09	0.06	1.4	0.8
Site (S)			0.000	0.000	0.000	0.000	0.004	0.003
Biochar (B)			0.000	0.000	0.000	0.000	0.000	0.000
Poultry manure (PM)			0.000	0.000	0.000	0.000	0.000	0.000
S × B			0.401	0.063	0.958	0.342	1.000	0.940
S × PM			0.751	0.100	0.909	0.255	0.991	0.915
B × PM			0.000	0.000	0.000	0.000	0.000	0.000
S × B × PM			0.275	0.784	1.000	0.789	1.000	0.976

Table 6
Results of multiple regressions of leaf area and tuber yield using soil physical properties.

Leaf area and tuber yield	R^2	Soil physical properties			p-value		
Leaf area	0.869	BD,	PO,	MC,	>0.000,	>0.000,	>0.005
Tuber yield	0.775	BD,	PO,	MC,	>0.002,	>0.001,	>0.000

*Significant at p = 0.05 level; BD: bulk density; PO: porosity; MC: moisture content.

Table 7
Results of multiple regressions of leaf area and tuber yield using soil chemical properties.

Leaf area and tuber yield	R^2	Soil chemical properties							p-value						
Leaf area	0.984	pH,	OC,	N,	P,	K	Ca,	Mg	>0.000,	>0.027,	>0.035,	>0.000,	>0.038,	>0.000,	>0.027
Tuber yield	0.987	pH,	OC,	N,	P,	K,	Ca,	Mg	>0.000,	>0.036,	>0.004,	>0.000,	>0.000,	>0.000,	>0.000

*Significant at p = 0.05 level; OC: organic carbon; N: total nitrogen; P: available phosphorous; K: exchangeable potassium; Ca: exchangeable calcium; Mg: exchangeable magnesium.

(2016) from their short-term and laboratory studies. This was due its high porosity and surface functionality compared to other types of soil organic matter.

The soil pH, OC, P, K, Ca and Mg concentrations increased with application of biochar and poultry manure. The increase in soil pH observed in biochar or poultry manure alone and their combination at different rates of application could be attributed to increased availability of organic matter and high concentrations of alkali metals and exchangeable basic cations (Ca, Mg, K and Na) present in their ash fractions which act as liming agents in acidic soils. The soil nutrient concentrations increased with poultry manure application, which was due to the nutrients release from the poultry manure after decomposition. Poultry manure has been demonstrated to increase soil OC, N, P, K, Ca and Mg (Kingery et al., 1994; Agbede and Ojeniyi, 2009; Adekiya et al., 2019). As the organic matter components of the poultry manure decomposed, nutrients were released to the soil. The increase in soil OC and nutrients with rates of application in plots amended with biochar compared with the control, could be attributed to various mechanisms and processes; the addition of nutrients contained in biochar, reduced nutrient leaching, improved nutrient retention, modified soil microbial dynamics and increased decomposition of organic materials in soil as reported in previous studies (Lehmann et al., 2003; Sohi et al., 2009). Major et al. (2009) demonstrated that biochar has the ability to retain nutrients contained in water solution held in its micropores by capillary forces. According to the authors, biochar particles are assumed to act like clay and thus hold large amounts of immobile water even at increased matric potentials. As a result of this, nutrients dissolved in this immobile water would be kept near the soil surface and would be available for plants (Major et al., 2009). In addition, Major et al. (2009) found that leaching of soil nutrients is reduced through adsorption of cations and anions to the biochar surfaces. The mechanisms responsible for increasing plant nutrient availability in biochar amended plots are increase in soil pH (in acidic soils), high OC content, high stability, nutrient retention (due to increase in cation exchange capacity and surface area) or directly release of nutrients from the biochar surfaces (Clough et al., 2013; DeLuca et al., 2015; Subedi et al., 2016).

The finding that optimum soil chemical properties (pH, OC, N. P, K, Ca and Mg) was attained at 30.0 t ha^{-1} biochar in combination with 10.0 t ha^{-1} poultry manure could be adduced to increased microbial activities and mineralization of nutrients induced by biochar and poultry manure addition, which should have increased nutrient availability. This treatment also added the highest amount of nutrients to the soil.

The results showed profound improvement in nutrient status and yield of sweet potato with poultry manure application. This is consistent with previous studies (Agbede et al., 2017; Adekiya et al., 2019). Poultry manure was found to contain plant nutrients including N, P and K as well as secondary and trace elements (Stephenson et al., 1990; Kingery et al., 1994). Enhancement of sweet potato productivity and nutrient status by poultry manure in this study was attributable to its low C:N ratio (7.72). The high nutrient concentrations and the low C:N ratio of the poultry manure in the current study should have hastened/increased decomposition and nutrient release for a crop like sweet potato.

The sweet potato growth and yield increased with the biochar application rates, which could be adduced to its liming effect in acidic soils, nutrient availability, and modification of physical soil properties that leads to increased root growth and yield (Partey et al., 2014; Nelissen et al., 2015; Mensah and Frimpong, 2018). The good performance of sweet potato due to biochar application could also be due to its continued persistence in the soil to ensure availability of nutrients.

The interactions between biochar and poultry manure were found to be significant for soil physical and chemical properties, growth and yield of sweet potato, indicating the potential of biochar in improving poultry manure use efficiency and enhancing better use of nutrients in the poultry manure. Biochar itself had low nutrient concentrations and decomposition rate because of the high C:N ratio of 65.5 (Table 2). However, the inclusion of biochar in such mixed treatment applications could have potentially reduced nutrient leaching and increased the nutrient holding capacity of the soil and thus increase in sweet potato tuber yield. In tropical soils, some researchers (Arif et al., 2012; Alburquerque et al., 2013; Agegnehu et al., 2016; Mensah and Frimpong, 2018) have demonstrated that the positive effects of biochar application on plant growth and yield are strongest when combined with organic manure or inorganic fertilizers. In an Ultisol in southern China, Peng et al. (2011) found that maize biomass was increased by 64% (without NPK fertilizer) to 146% (with NPK fertilizer) after biochar application (2.4 t ha^{-1}). The highest soil nutrient and sweet potato yield obtained when 30 t ha^{-1} biochar + 10 t ha^{-1} poultry manure were applied attest to the positive cumulative effect of biochar and poultry manure on soil productivity. Biochar and poultry manure had synergistic relations in terms of combination to further enhance soil OC and nutrients and improve soil physical properties. The present work shows that sweet potato yield is influenced by availability of soil OC, N, P, K, Ca and Mg. Sweet potato, generally has high demand for K, N, Ca, Mg and other trace elements. The deficiencies of these nutrients, as well as aluminium toxicity have been reported to limit growth and yield of root crops such as sweet potato (O'Sullivan et al., 1997). The value of soil K, N, Ca and Mg was significantly ($p = 0.05$) increased at different levels of biochar and poultry manure application. Potassium (K) had been reported to play a critical role in sweet potato production (O'Sullivan et al., 1997; Njoku et al., 2001). For sweet potato to thrive, one of the key nutrients it needs is K. It impacts the plant's ability to efficiently use nitrogen and aids water uptake. Potassium is also essential for improving nutrient value and optimizing yield.

The results of the multiple regressions analysis showed that bulk density, porosity and moisture content significantly influenced the tuber yield of sweet potato. The multiple regressions showed that pH, OC, N, P, K, Ca and Mg significantly influenced the tuber yield of sweet potato. Reducing bulk density and increasing porosity and moisture content, and increasing soil chemical properties were consistent with increase in tuber yield of sweet potato. Therefore sweet potato tuber yield in this study is dependent on bulk density, porosity and moisture content and soil chemical properties. Decreased/reduced soil bulk density is known to increase root penetration, prevent compaction and thereby enhancing water and nutrient uptake and root formation (Lampurlanes and Cantero-Martinez, 2003; Agbede, 2008) which will consequently increase growth and yield. Reduced bulk density and high porosity of biochar and poultry manure amended soils and the consequent reduced mechanical impedance for sweet potato root growth and this resulted in increased length and size of the sweet potato tubers. The highest sweet potato yield obtained when 30 t ha^{-1} biochar + 10 t ha^{-1} poultry manure were applied was attributable to the better improvement in soil physical and chemical properties of this treatment (Adekiya et al., 2019).

5. Conclusions

Sandy soil and sandy loam soil of the humid tropics amended with sole biochar and poultry manure and or their combination at different levels improved soil physical (reduced bulk density, and increased porosity and moisture content) and chemical (pH, OC, N, P, K, Ca and Mg) properties, growth and tuber yield of sweet potato. The improvement in growth and tuber yields of sweet potato was attributed to improved soil physical and chemical properties, indicating the ability of the biochar in improving poultry manure use efficiency and enhancing better use of nutrients in the poultry manure. The highest dosage of combination of 30.0 t ha^{-1} biochar and 10.0 t ha^{-1} poultry manure gave the best improvement in soil properties and sweet potato performance. This treatment is therefore recommended for soil fertility management and sweet potato production in the study areas (rainforest agroecology of southwest Nigeria). However, there is the need therefore to evaluate these findings using other agroecological zones, soils and crops, and in addition to variable application rates and combinations of biochar or poultry manure. Long-term studies focusing on the persistence of biochar effects on soils and crops are recommended.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability statement

All data generated or analysed during this study are included in this manuscript.

Acknowledgements

The authors acknowledge the financial support provided by the Tertiary Education Trust Fund, Nigeria (TETFUND/DRSS/POLY/OWO/2016/RP/VOL.1: 2016–2017 TETFund Research Projects (RP) Intervention: Release of Fund 6th Batch (RP Disbursement) and the Management of Rufus Giwa Polytechnic, Owo, Ondo State who recommended the research work for the TETFund for funding.

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